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Research article

Interactive simulation and visual analysis of social media event dynamics with LLM-based multi-agent modeling

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ABSTRACT

With the increasing role of social media in information dissemination, effectively simulating and analyzing public event dynamics has become a key research focus. We present an interactive visual analysis system for simulating social media events using multi-agent models powered by large language models (LLMs). By modeling agents with diverse characteristics, the system explores how agents perceive information, adjust their emotions and stances, provide feedback, and influence the trajectory of events. The system integrates real-time interactive simulation with multi-perspective visualization, enabling users to investigate event trajectories and key influencing factors under varied configurations. Theoretical work standardizes agent attributes and interaction mechanisms, supporting realistic simulation of social media behaviors. Evaluation through indicators and case studies demonstrates the system's effectiveness and adaptability, offering a novel tool for public event analysis across open social platforms.

1. Introduction

Social media has become an important platform for the dissemination of major events and the discussion of hot topics due to its real-time, open characteristics and extensive user participation. The complex behaviors of users on social media platforms generate diverse data. The rapid dissemination of multidimensional information, including text content, interactive relationships, and geographic locations, promotes the diffusion of opinions, affects the evolution of group emotions, and even the development of real events. The diverse dissemination of information has led to the collision of multiple opinions, and also brought about the differentiation of public positions and the uncertainty of the development of public opinion.

As social media platforms continue to expand their influence in information dissemination, gaining a deeper understanding of the dynamic evolution of public opinion events and their potential impacts has become a critical research topic. Social simulation is one of the fundamental tools for analyzing this process. By driving model with appropriate input and observing the corresponding output, researchers can explore possible social behaviors (Bratley et al., 2011). *Simulation for the Social Scientist* (Gilbert and Troitzsch, 2005) identifies two key elements of social simulation from social science perspective: the interaction and dynamic feedback among individuals, and the changes in the state of group. Social simulation can thus be conducted through both macro and micro approaches. Macro-level simulation models the evolution of group states based on the whole system, focusing on the overall dynamics of group state transitions. In contrast, micro-level simulation is based on individual agents, where interaction behaviors are constrained by predefined rules or parameterized models.

With the continuous advancement of artificial intelligence technology, the depth and usability of agentbased modeling and simulation have been significantly enhanced. By integrating human–computer interaction techniques, researchers can establish virtual experimental environments, freely and dynamically adjust the behaviors of key participants or implement strategic interventions. As one of the major advancements in the field of deep learning, large language models, trained on vast corpora, have demonstrated exceptional natural language understanding and generation capabilities (Brown et al., 2020), which make them an ideal core for simulating human-like intelligent agents: they can perceive and comprehend the environment through textual descriptions, execute reasoning, decision-making and action based on given task requirements, and achieve effective interactions with other agents or the external environment through textual outputs.

Beyond generating high-quality content, large language models can adjust their generated outputs according to predefined

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rules, exhibiting a considerable degree of controllable generation, which further enhances researchers' precision in regulating agent behaviors. In the context of social media, researchers can leverage large language models to simulate users within social networks, enabling diverse roles and interaction patterns. They can dynamically adjust user behavior preferences, emotional attitudes, and responses to information, generating actions and interactions in accordance with predefined rules. This leads to more realistic and observable social behaviors, thereby improving the understanding of individual behaviors, uncovering interaction mechanisms among users, and elucidating the influence on the overall dynamics of public opinion events.

In this paper, we aim to introduce large language model-driven agents as roles with certain social attributes into the discussion of public opinion events. Through interactive visual analysis, we explore how agents with varying characteristics process information, adjust emotions and stances, provide feedback, and influence the development dynamics of events. The contributions are as follows:

- **Agent-Based Modeling in Social Media Simulation**: We systematically incorporate LLM-driven agents into public opinion event simulations, equipping them with customizable individual attributes such as personality, stance, emotions, and engagement levels. By constraining agents' perception of information and their subsequent responses based on these characteristics, the simulation better reflects the diverse behaviors observed in real-world social networks. Additionally, group interaction mechanisms enhance the accuracy and realism of modeling complex social dynamics. We visualize each stage of the simulation, enabling in-depth analysis of opinion dynamics.
- **An Interactive Social Media Simulation Framework**: We propose a simulation framework supporting interactive mechanism and enabling dynamic adjustment of agent quantities, characteristics, and behavioral rules. It also allows modifications to environmental factors, such as adjusting the initial scale or setting opinion biases, to simulate various event evolution scenarios. The results can be iteratively refined for parameter refinement. This flexibility allows researchers to intervene in the simulation process, observe the real-time impacts of agent behaviors, and identify key influencers.
- **A Visual Analysis System**: By integrating visual analysis with human–computer interaction techniques, we design and implement a visual analysis system that supports multi-agent simulation. The system provides multiple functional interfaces, allowing users to directly influence various stages of the simulation and visually present information such as the timeline, information propagation path, and the dynamic evolution of emotion and stance. The system combines dynamic simulation with interactive analysis, offering real-time feedback on how different parameters and settings affect the simulation results. We provides an effective technical tool and methodology for the simulation and analysis of public opinion events. The effectiveness of the research methods and system is demonstrated through several case studies.

2. Related works

This section reviews related work on social media visual analysis, LLM-based agents and social simulation.

2.1. Social media visual analysis

With the rapid development of public social media, various platforms have accumulated large amounts of data with rich features. Researchers analyze sentiment, stance, and event development based on the semantics, images, and other information published on social media platforms. DemographicVis (Dou et al., 2015) collects detailed demographic information and supports interactive analysis of groups with well-defined characteristics. WeiboEvents (Ren et al., 2014) uses tree, circular, and sail layouts to display various aspects of information diffusion. D-Map (Chen et al., 2016) uses a map metaphor to illustrate how information diffuses between communities, centered around individuals.

Content is a key component of social media, where users generate various forms of content to share information and express opinions. Visual BackChannel (Dörk et al., 2010) is one of the earliest systems to visualize social media content, using a timeline to display keywords and topics, a circular view to represent participating users, a text list to show raw data and images, and interactive search and filtering options. Word clouds (Viegas et al., 2009) represent the frequency and importance of keywords through their size in the interface. SentenTree (Hu et al., 2016) visualizes the co-occurrence patterns of high-frequency keywords within sentence-level information. RoseRiver (Cui et al., 2014) uses a tree-cutting algorithm to detect topic hierarchies over different time periods, visualized through a river metaphor. Krueger et al. (2014) further summarize the needs of users in sentiment analysis, providing a comprehensive display of how sentiment and attitudes change over time. This study focuses on the multidimensional information of social media data, and based on existing work, integrates the requirements of simulation analysis to implement a multi-view interactive visual analysis system.

2.2. LLM-based agents

The field of artificial intelligence is dedicated to developing artificial entities with human-like intelligence and capabilities. The rise of large language models has expanded the boundaries of agent capabilities, and combined with multi-modal perception technologies and tool invocation, the perception and action spaces of agents have been extended, exhibiting more human-like behavioral characteristics (Xi et al., 2025). In more basic applications, agents follow user instructions to complete tasks such as goal decomposition and task planning (Huang et al., 2022). To accomplish more complex and realistic tasks, the basic characteristics of roles are incorporated into natural language prompts, allowing agents to think based on the fundamental information of their roles. Character-LLM (Shao et al., 2023) leverages encyclopedic information to obtain personal profiles of roles and trains models on this foundation to enable highly credible and authentic interactions. In the work by Zhang et al. (2023), agents are given different traits to form a diverse society, with confident agents being more persuasive in social collaboration, while another type is objective and impartial, making them easier to persuade. In Hu et al.'s work (Hu et al., 2025), individuals in rumor propagation networks are assigned names, ages, professions, and personality traits, resulting in differentiated acceptance of rumors and propagation intentions, driving the spread and evolution of false information within a group.

Building on the incorporation of personality traits into agents, one of the key areas of research is how to regulate the agents' perception and action modes to achieve efficient interaction with the outside world. In ReConcile (Chen et al., 2023), agents communicate through a round-table format, with each agent attempting to persuade others and improve their own answers, while maintaining confidence to represent their stance. Richelieu (Guan et al.,

2025) restricts agent activities to a diplomatic context, where agents use reasoning to form beliefs about social relations and intentions, reflect on proposed sub-goals, negotiate with others, and take diplomatic actions. S^3(Gao et al., 2023) establishes a social network simulation system from three perspectives: emotional attitudes, interaction behaviors, and social network dynamics, where agents based on large models realistically and reasonably simulate user behaviors, displaying group behaviors.

In summary, prior works have focused on enriching agents with diverse personalities, social traits, and reasoning abilities to better mimic human behaviors in specific contexts. Building on this, our work combines large language models with a flexible multi-agent simulation framework, allowing agents to dynamically adjust their stance, emotion, and interaction behaviors in evolving social media events. This integration enables us to simulate nuanced opinion evolution and collective dynamics in an interactive visual analysis system.

2.3. Social simulation

Social simulation drives system models with appropriate inputs and observes the corresponding outputs, primarily exploring possible social behaviors that may arise within the system. Agent-based modeling and simulation allow researchers to explore the behavior of individuals within large systems. Although such models cannot make precise predictions or fully reflect social realities, they enable researchers to identify social mechanisms by examining how local interactions influence global patterns (Törnberg et al., 2023). Large language models enhance the agents' reasoning, execution, and interaction capabilities. To further enhance the depth and usability of simulations, user interactions have been integrated into the social simulation process, allowing users to adjust parameters in real-time and guide agent behavior (Zhang et al., 2024). Human–computer interaction improves the accuracy of simulations, making the research more user-centered.

Horton (2023) assigns agents endowments, information, and preferences through prompts, replacing human participation in economic behavior simulations. Park et al. (2023) proposed a method with an effective cognitive architecture that supports decision-making, long-term planning, and higher-level reflection, to simulate the lives of 25 agents over two days in a virtual town. In the system built for this study, users can define, control, and intervene in the agents and environment using natural language instructions. Memory Sandbox (Huang et al., 2023) allows users to directly manipulate agents' memories to achieve cognitive alignment. Social Simulacra (Park et al., 2022) utilizes the capabilities of large language models to generate social interaction behaviors of agents, including posting, replying, and anti-social behaviors. Agents successfully replicate and mimic real-world phenomena in simulated environments.

Although there has been some application and research on large language model-driven agents in social media simulation, this study focuses on how to assist simulation and result analysis through visualization technologies, building an interactive simulation-supporting visual analysis system by combining theoretical research with practice.

3. Framework

Based on an in-depth investigation of the research background and recent advances in LLM-driven agent modeling and social simulation, we propose a flexible simulation and visual analysis framework to explore the dynamics of social media events. This framework bridges the gap between realistic agent cognition, emotion and stance evolution, and interactive scenario analysis.

This section presents an overview of our research, including the data used, research tasks and requirements, and proposes the workflow followed by the visual analysis system.

3.1. Data description

The data used in this study comes from Sina Weibo, a well-known social media platform in China. With a large user base, high interactivity, and rich content, it serves as an important data source for social media research and analysis. The proposed methodology and analytical framework are highly generalizable and can be applied to other public social media platforms.

One of the defining features of social media platforms like Sina Weibo is that they allow users to emphasize the themes of their posts using in-text tags (e.g., "#topic tag"). Tags indicate the events to which the posts belong. For posts without explicit tags, we focus on in-text keywords to identify and associate them with relevant events based on their textual content. During the data collection process, we first manually select popular keywords and topic tags related to specific events to construct a keyword dictionary. We then employ web crawler to retrieve posts containing these keywords or tags, along with their associated reposts.

Each collected data entry includes the following key attributes: timestamp, post ID, username and user ID, textual content. Based on this, a large language model is employed to annotate the sentiment and stance information, followed by manual verification. During the subsequent data cleaning phase, empty texts and other irrelevant data are removed to construct the foundational dataset. The generated dataset is stored in JSON format, with each record containing essential post information along with sentiment and stance labels, as well as sentiment intensity.

For example, in the "Gauze Gate" case study, the dataset includes 67 original posts and 9559 reposts related to the event. After data cleaning, a total of 3615 valid entries were retained for subsequent analysis and agent-based simulation. This dataset provides a structured and semantically reliable foundation for simulating social media event dynamics using LLM-driven multi-agent modeling.

3.2. Research tasks

Based on an in-depth investigation of the research background and related work, we first conduct a detailed analysis of the research tasks, providing a theoretical foundation and framework for the subsequent system design. By deconstructing the entire simulation analysis process, the research tasks can be divided into three parts:

T1: Social Media Data Analysis and Simulation Environment Construction At this stage, the collected social media data must first be thoroughly analyzed to ensure a comprehensive understanding of its structure and characteristics. Based on this, a simulation environment is then constructed. By leveraging visual analysis techniques, various dimensions of the data — such as temporal patterns, sentiment distribution, topic popularity, and user behavior — can be intuitively presented. Additionally, interactive exploration is supported, allowing for in-depth examinations of different aspects of the data.

T2: Environment and Agent Configuration & Simulation Implementation At this stage, the dynamic simulation of social media is achieved by adjusting various configurations within the simulation environment, such as the number of agents, individual attributes, and the initial scale of posts. Agents take actions according to predefined rules, generating corresponding feedback. Interactions occur at multiple levels, including agent-to-agent interactions, agent–environment interactions, and agent-external event interactions. The data generated throughout the simulation process provides valuable insights into the evolution of public opinion events on social media.

T3: Visual Analysis and Interactive Adjustment The core of this stage is to conduct visual analysis based on the simulation environment and results from the previous stages. The system intuitively presents the overall evolution of the simulation process, allowing users to adjust simulation settings based on the results — such as modifying agent deployment strategies or altering external events — and compare multiple simulation outcomes. Through an interactive iterative process, users can explore the impact of different configurations on the simulation results, analyze how agent behaviors drive event evolution, and further enhance the accuracy and applicability of the simulation.

3.3. Design requirements

Based on the analysis of research tasks, the system should meet the following key requirements to construct a realistic social media simulation environment and enable targeted validation and analysis in experiments.

R1: Simulation Initialization The initialization of the simulation process consists of two core components: environment configuration and agent deployment. Based on the original data, the environment settings allow for adjusting the probability of posts with specific sentiment and stance inclinations being included in the environment, thereby establishing an initial discussion atmosphere with particular biases. Sudden public events are introduced through mandatory broadcasts, ensuring that all agents receive the relevant information. To prioritize the processing, such events are assigned a high weight. In real-world scenarios, different types of users play diverse roles in information dissemination. Therefore, the system must be designed to accommodate and implement configurations that reflect the characteristics of these roles.

R2: Visual Analysis Before the simulation, an analysis of the original data is conducted to establish a foundational understanding, providing a basis for simulation configuration. After the simulation, a comparative analysis is performed to examine how results vary under different environments and configurations. The analysis focuses on several key aspects, including content, semantic information, reposting networks, individual behavioral patterns, and state transitions.

R3: Interactive Iteration The simulation results need to be prominently displayed, highlighting key content to help users understand the dynamic changes within the context of the original data. Building on the analysis of current results, the system supports modifications to the simulation configuration and further reuses part of existing outcomes. This enables more efficient iterative analysis and optimization.

3.4. Workflow

Taking into account the research tasks and requirements, our visual analysis system follows the workflow shown in Fig. 1. To enhance clarity, we add brief explanations for each module's role: the initialization module prepares environmental data and agent attributes; the execution module simulates agent interactions and records dynamic results; the visual analysis module explores the simulation outputs to uncover underlying social media event patterns and supports iterative refinement.

The initialization of the simulation includes the configuration of both environmental data and agent data. Posts undergo cleaning and annotation. Event scale and biases are configured through the system interface, collectively forming the external environment for agent interactions. The initialization of agents involves defining their attribute features and interaction rules. Each agent is assigned unique personality, initial emotion and stance at the beginning of the simulation, and dynamically evolves throughout the simulation.

The execution stage, refers to the core simulation module that runs the agent-based social media event dynamics. During simulation execution, agents perceive information from the environment, adjust their emotions and stances, and influence the environment through behaviors such as reposting. The simulation results are recorded in a structured format, including trajectories of agents' emotion and stance changes, information dissemination paths, and the evolution of event discussion intensity.

The visual analysis phase focuses on the multidimensional information in the simulation results to gain a deeper understanding of the event evolution process and its underlying mechanisms. The analysis includes group and individual sentiment and stance changes, transformations in reposting structures, semantic shifts, and fluctuations in discussion intensity. The interim findings from exploratory analysis can inform adjustments to simulation strategies. The system allows users to modify configurations based on existing result and rerun the simulation, enabling an iterative process between simulation execution and visual analysis, thereby refining and completing the study.

4. Methodology

This section introduces the proposed interactive simulation methodology, covering agent characteristic, interaction mechanism, and content generation. Agent characteristic part discusses the classification of agents and the configuration of their intrinsic attributes, where agents with different characteristics follow distinct behavioral patterns during the simulation. Interaction mechanism part details the complete process of agent perception, action, and feedback within the simulation. Content generation part leverages the natural language generation capabilities of large language models, with generated data recorded in a structured format for subsequent analysis.

4.1. Agent characteristics

To realistically model social media dynamics, agent characteristics and behaviors are designed based on established theories in social psychology and online behavior modeling. Prior research indicates that user stance, emotional response, and engagement levels are key factors shaping information diffusion and public opinion formation in online communities (Gao et al., 2023; Deffuant et al., 2000; Jager and Amblard, 2005; DeGroot, 1974; Hunter et al., 2014; Rainer and Krause, 2002). Therefore, our framework incorporates these core dimensions to enable flexible configuration and dynamic simulation of agent interactions.

Agents are primarily categorized into two types: LLM-driven agents and rule-based agents. LLM-driven agents play a pivotal role in the simulation and are further divided into two subtypes based on their expressive tendencies: opinion publishers, who actively express viewpoints, and opinion receivers, who are more inclined to adopt others' perspectives. The former simulates opinion leaders on social media, shaping event dynamics by disseminating content with specific stance, while the latter represents general social media users, who primarily take information and respond based on their preexisting views. Rule-based agents rely more on the emotional atmosphere to determine their behavior. They are highly influenced by surrounding interactions, making them a more perceptive user group.

Throughout the simulation, agents continuously adjust their state in response to external stimuli and internal logic. This dynamic adaptability enhances their realism, making their behavior more closely resemble that of real social media users. By capturing these evolving interactions, the simulation more accurately reflects the mechanisms of public opinion formation.

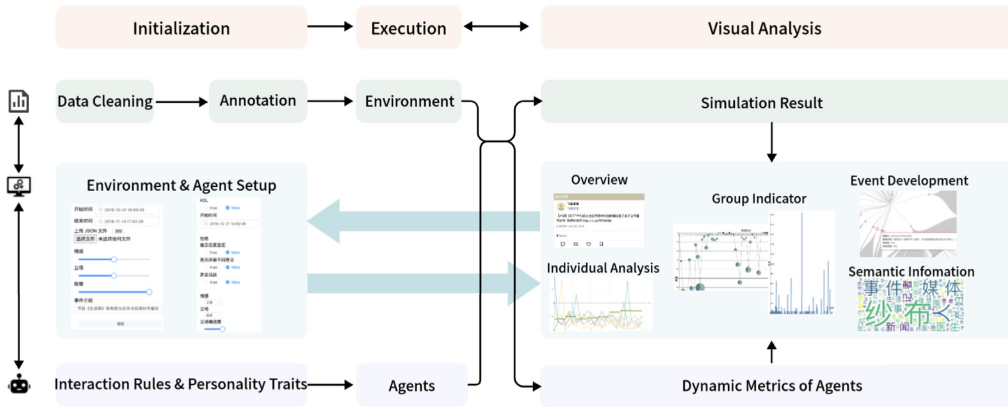

Fig. 1. Overall workflow of the system, including data preprocessing, agent and environment configuration, simulation execution, and visual analysis.

We focus more on personality traits that influence expression styles and external social behaviors. Based on theoretical research (Deffuant et al., 2000; Jager and Amblard, 2005), three main characteristics are identified: attitude firmness, response to opposing views, and activity level.

Throughout the following discussion, the term stance denotes the position an agent holds towards a topic (e.g., supportive, neutral, or opposing), whereas state indicates the agent's internal emotion and stance, which may dynamically affect its behavior.

Attitude firmness determines the difficulty of an agent changing its stance when receiving external information. During participation in social media activities, agents accept new information and update their state based on a comprehensive consideration of their own conditions. Agents with firm attitude have a higher threshold for stance change and are less likely to alter their position due to environmental changes.

Response to opposing views determines how an agent browses posts. In addition to the influence of popularity, the probability of an agent browsing a post is affected by social relationship, as agents tend to interact more with individuals who share similar viewpoints. Agents with the opinion-blocking attribute impose stricter filtering measure on posts and users whose viewpoints differ significantly from their own.

Activity level determines an agent's willingness to post and the number of posts it makes. It is generally believed that individuals with greater emotional or stance fluctuation have a stronger inclination to express themselves. Based on this, we distinguishes individual differences among agents. With the same level of fluctuation, active agents have a higher probability and frequency of posting, showing a greater level of participation.

During the simulation process, the behavior of agents is constrained by above characteristics. We effectively achieve the diversity of participating individuals.

4.2. Interaction method

Agents perceive environmental information through text content, updating their emotion and stance accordingly. This process is influenced by several factors, including the understanding of the event, the receptiveness to new information, and the intensity of emotion. They adjust based on the content of posts in the environment, the popularity of the posts, and the external emotional atmosphere. They then publish information based on the updated state.

To simulate real browsing patterns, agents acquire information with a certain probability based on the post's popularity and relevance. Posts with higher popularity are more likely to be pushed, allowing the agent to prioritize important information. At the same time, agents do not directly interact with other agents, but by evaluating the content posted by other agents, they can gauge the degree of agreement with those posts. This degree of agreement is closely related to the post's likelihood of being pushed, simulating the attention relationship formed through indirect interactions among social media users in reality.

All posts in the current environment are arranged in chronological order. A specific number of posts (depending on the event and data scale) are grouped into a time slice. Within each time slice, the agents take actions sequentially.

The emotion and stance infomation of all posts within the current time slice are aggregated as environmental feedback and are prioritized for delivery to opinion leader users. These agents, based on their pre-defined stance and the event's development, publish information, and the posts are added to the next time slice, where they will have a higher push probability when browsed by other agents. In the next stage, rule-based agents take actions. They rapidly browse social media information based on specific rules, focusing on the changes in emotion and stance indicators. In the final stage, regular users take actions, which can be further divided into browsing posts, reading emotion and stance, taking blocking actions, updating emotion and stance, and expressing opinions.

In each time slice, agents randomly browse a certain number of posts (at least one). Posts with higher popularity or those published by users with similar stances to the agent are more likely to be browsed. Agents can only browse posts in the current time slice, and posts browsed in previous time slices are retained in memory. The memory module allows agents to adjust their behavior based on their interaction history. If the stance of a post the agent browses differs too greatly from its own stance, the agent may block all other posts from the user who made the post, with the blocking mechanism becoming effective in the next time slice. The updating of emotion indicators mainly relies on the large language model's understanding ability. In addition to qualitative indicators such as positive or negative, an intensity quantitative indicator is used to refine emotion. The update rules for the stance vary depending on whether the agent is stance firm or not:

For agents with a firm attitude, the rules for stance updates are more complex. The agent will only further process information if its strength exceeds a certain threshold. Otherwise, only the stance confidence will be randomly disturbed. Furthermore, if the stance of the information matches the agent's stance, agent's confidence slightly increases, consolidating the agent's stance. If the stance of the information is different, and if the information strength exceeds a change threshold (slightly higher than the previous threshold), the agent's stance will change, and the confidence will decrease due to the stance change. Otherwise, the confidence will only slightly decrease without changing the stance.

For agents with an uncertain attitude, the rules are simplified. If the information strength exceeds the threshold, the agent's stance aligns with the information's stance, and the confidence is proportional to the information strength. If the information strength does not exceed the threshold, only the confidence is slightly adjusted, but the stance remains unchanged.

4.3. Content generation

The text generation of agents relies on large language models. Agents can thus generate text that aligns with their personality and context based on their current emotional state and stance. The system conveys key elements such as emotional state, stance information, and contextual background to the model via natural language prompts, driving the model to generate text. Agents can adapt flexibly to changes in the environment, participate in the evolution of events and influence the development of those events through expressing opinions.

Data generation is done on an agent-by-agent basis. For each time slice (of non-uniform duration), the behavior of each agent is stored as a data entry: the posts browsed by the agent are recorded by ID, with stance and emotion recorded as categorical variables. After browsing a post, the agent's confidence level and emotional intensity are updated and recorded as numerical values. Possible shielding behaviors are recorded by the user ID. The format of the agent's post follows the same structure as the original post data.

To control the consistency between generated content and the simulation state, we design adaptive prompts that incorporate explicit constraints on sentiment polarity, emotional intensity, and stance alignment. For example, the prompt dynamically encodes the agent's current emotion score and stance tendency to guide text generation towards coherent outputs. We also integrate sampling strategies to mitigate issues such as repetitiveness and generic phrasing, ensuring greater linguistic diversity and stylistic variation in the generated posts.

We employ Qwen2.5-3B-Instruct and GPT-3.5 as the core text generation engine. Prompts are dynamically constructed to reflect each agent's current emotion, stance, and context. For example, a prompt may explicitly encode an agent's sentiment score and target stance shift, ensuring that generated content aligns with the desired behavioral trajectory. One example prompt template is provided in Appendix.

5. System

Based on the above analysis, we develop an visual analysis system. The system supports the aforementioned functions through multiple interfaces. This section introduces the system design and implementation from three aspects: data display, simulation setup, and result analysis. The overall system view is shown in Fig. 2.

5.1. Data display

The system displays the original post data and provides an overview of the basic event information, helping users gain an understanding and conduct preliminary analysis of the event to build a foundational cognition of the current event.

On the left side, **time selection bar**(Fig. 2 A) is provided to filter posts within a specific time range. Below, the **thematic content**(B) and **post texts**(C) are displayed to provide detailed information about the event. An **input box**(D) is also located below, allowing users to search through all posts using keywords.

The **timeline**(E) is represented as a bar chart, displaying the overall trend of discussion heat over time during the event. Users can hover over any bar element to view specific post details. By right-clicking on any point in timeline and entering information, a sudden news can be added to the list and highlighted. Users can brush-select the timeline to update the range of data displayed and analyzed.

The system modularizes functionality through **checkbox**(F), where the selected functional interface is displayed below. This design ensures that the necessary analysis functions are implemented while maintaining a clear interface and reasonable layout.

5.2. Simulation setup

Apart from external factors such as policy announcements and sudden hot news, which are set on the timeline, the **environment settings** component allows users to configure the start and end times of the simulation, which define the observation window for simulating the timeline of event propagation and agent interactions. This temporal setting aligns the simulation period with the real-world event duration and affects how agents perceive, respond, and repost over time. Additionally, users can reuse previous simulation results by loading files. The system also supports setting a brief description of the event, stored as natural language text, which is provided to agents during the simulation as their initial understanding of the event. Through parameterized settings, users can further filter the source data within the limited simulation time and adjust the probability of posts being selected based on their inclination. After clicking "Save", the foundational data for the simulation — i.e., the simulation environment — is stored, and the final number of selected posts is displayed below. With these settings, users can modify the overall scale of the event and adjust the emotion and stance bias, thereby creating different event backgrounds for the simulation.

Agent settings are based on the theoretical analysis mentioned earlier. The interface allows users to configure when agents join the discussion and begin influencing the environment, as well as define their attributes and personalities. These settings include whether the agent is an opinion leader (whose posts are more likely to be viewed and carry greater weight in interactions), whether the agent has a firm stance (requiring stronger information stimuli to change their views), whether the agent filters out opposing opinions (determining how they handle posts and users with differing views), and whether the agent is an active user (which affects posting frequency). To mimic real social media users, each agent has a defined emotion, stance, and confidence level regarding their stance upon joining.

Once all settings are confirmed, the current agent is added to the list, where users can scroll to browse agents or click to delete a specific agent. By clicking the "Simulate" button, the agent and environment settings are sent to the backend to initiate the simulation process.

5.3. Result analysis

The system employs multiple views to analyze simulation results from various perspectives, following the key points of visual analysis outlined in the previous requirements analysis.

Similar to the timeline, a **histogram** (Fig. 3a) displays the changes in post popularity over time. . Posts are arranged sequentially, with the horizontal axis encoding their order rather than specific timestamps. Users can switch between sorting posts by time or popularity via a button, allowing them to focus on high-engagement core information. To adjust information density, reposted content is hidden by default but can be included at the

Fig. 2. The system overview. The figure shows the system framework and some functional interfaces: (A) Time Selection (B) Topic Content (C) Post Content (D) Search Input Box (E) Timeline (F) Function Selection Checkbox (G) Environment Settings (H) Agent Settings (I) Group Emotion and Stance.

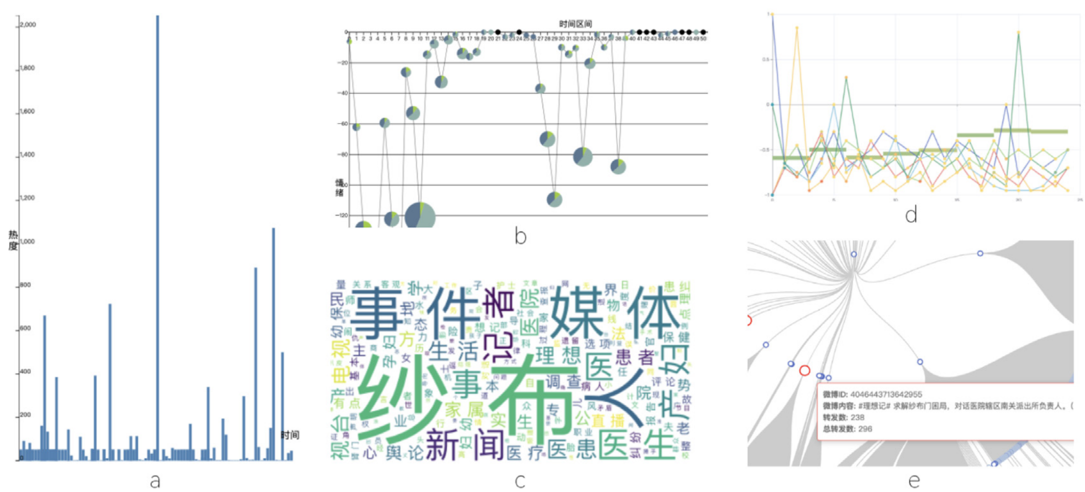

Fig. 3. Result analysis.

user's discretion. Additionally, a minimum popularity threshold can be set to filter the displayed posts.

The **group emotion and stance** (Fig. 3b) component integrates line charts and pie charts to visualize changes in emotion, stance, and discussion over time. The emotion indicator for a given time period is obtained by summing all post weighted by their information strength. Each pie chart element encodes time on the horizontal axis and emotion on the vertical axis, with positive and negative values representing positive and negative sentiments, respectively. The radius encodes the overall discussion scale in that time range, while the pie chart's area represents the proportion of users holding different stances. Users can adjust the time range to any whole number of hours or days to analyze data at different temporal granularity.

Word cloud (Fig. 3c) is used to analyze the semantic information of the posts, helping to reveal important keywords and their relative weights within the text data, highlighting the themes of the discussion. The font size of each word is directly proportional to its frequency of appearance in the text.

After obtaining the simulation results, the **emotional and stance changes of agents** (Fig. 3d) throughout the process also need to be closely monitored and analyzed. All agent data is visualized as a line chart, with the horizontal axis representing time and the vertical axis representing sentiment, with node colors indicating stance changes. Hovering over a node will display information for that time slice, including agent ID, sentiment indicators, stance indicators, and the posts that the agent has browsed during that time slice. Additionally, simulated data generated by rule-based agents is incorporated into the visualization. The group indicators for these agents are presented using square elements. The horizontal axis represents individual time slices, while the vertical axis encodes group emotion bias. The color of the element shows the group's stance bias.

In our system, each original post can generate multiple reposts by other agents, forming a branching structure that represents the flow of information across the network. This structure is visualized as a repost tree, where the parent node is the original post, and each branch indicates a repost chain. The **repost tree** (Fig. 3e) visualizes the hierarchical structure of message diffusion during the event propagation process. The radius of each node encodes its forwarding level. A virtual node is used as the root node at the center of the view, with each layer of posts arranged

Table 1
Evaluation Result.

Time slice	$\Delta Bias_s$	ΔDiv_s	$\Delta Bias_e$	ΔDiv_e
1	−0.0772	0.2137	0.0664	−1.0635
2	−0.0337	0.0814	0.1605	−0.9079
3	−0.0006	−0.0039	0.1586	−0.9195
4	−0.0826	0.1158	−0.0532	−0.9153
5	−0.0505	0.1099	−0.1402	−0.8156
6	0.0034	0.0134	−0.2581	−1.1416
7	−0.0072	0.0266	−0.3722	−0.9372
8	0.0011	0.1362	−0.5212	−0.9639

outward. After the simulation ends, the data is reloaded, and posts published by agents during the simulation are highlighted within the tree. Moreover, posts browsed by agents are highlighted in the retweet tree, helping to identify key turning points in sentiment and stance.

6. Evaluation

This section first evaluates the effectiveness and rationality of the large language model-driven agent system in simulating social media environments. It then demonstrates the feasibility and scalability of the proposed research methodology through case studies on different datasets.

6.1. Reasonableness verification

We selects the core discussion time range based on popularity. In this environment, multiple agents with different initial emotions and stances are deployed, ensuring that the proportion of agents' emotions and stances aligns with the proportions observed in real data during the initial time slice.

Lorenz et al. (2021) proposed using bias ($Bias_s$) and diversity (Div_s) metrics to measure the static distribution of stances. Bias reflects the deviation of the average stance from neutral state, while diversity is represented by the standard deviation of stances. These metrics can similarly be extended to evaluate emotional distribution ($Bias_e$, Div_e). In each time slice, the mean and standard deviation of agents' states are calculated, and the differences between these values and those derived from the original data are computed. A comparison of simulation metrics and real-world metrics is presented in Table 1.

If the absolute error is less than 0.1, we consider the simulation results to have minimal deviation from the real data. The threshold of 0.1 for absolute error was selected based on prior literature on opinion dynamics model validation (Mou et al., 2024), where deviations below this value are commonly interpreted as indicating high fidelity between simulation and observed data. Absolute errors in the range of 0.1 to 0.3 are deemed acceptable, while errors exceeding 0.3 indicate significant deviation. In the table, absolute errors within the 0.1–0.3 range are marked with underlining, while those exceeding 0.3 are highlighted using wavy underlining. The results show that the error between the stance distribution of simulated agents and real-world stance distribution metrics mostly falls within the ±0.1 range, indicating a close alignment with actual data. Although agent emotion distribution appears slightly extreme in certain time slices, the overall mean absolute error is 0.2, suggesting that the deviation remains within a reasonable range.

Dynamic Time Warping (DTW) (Müller, 2007) and Pearson correlation coefficient (Cohen et al., 2009) are used to measure the similarity between the time series of average stance in the simulation and real data. The results are shown in Table 2. This indicates that the simulated data exhibits a high degree of temporal similarity to real data, demonstrating that the simulation results effectively capture the dynamic changes observed in the real world.

Table 2
Similarity Evaluation.

	DTW distance	Pearson correlation
indicator	0.1268	0.7144

6.2. Case 1: Gauze Scandal

The Gauze Scandal was an important social event that took place from October to November 2016. A TV programme first reported the medical accident of a woman with gauze in her body after giving birth, which triggered social criticism of the hospital and doctors. As public opinion spread and relevant institutions intervened in the investigation, the truth of the incident was reversed.

Our system generates a keyword word cloud based on the frequency of words. In the overall process, keywords such as "gauze", "media", "reporter", "hospital", "doctor", and "maternity" are highlighted. Combined with the analysis of group stance and emotion (Fig. 4), the discussion heat was low at the beginning of the incident, and the group stance was biased towards the patients. As public opinion reversed, the group sentiment quickly became more negative, and the stance shifted towards the hospital. The peak of emotional change and discussion heat occurred on November 7th and 10th, corresponding to CCTV's report on the incident and some bloggers' discussions on media professional ethics. The distribution of group emotions seemed to break around November 14th, as the group sentiment shifted from negative to neutral, and the discussion of the incident came to an end.

Based on this set of posts, simulation and subsequent analysis were conducted, with opinion leaders holding neutral emotional states and stances on the hospital side, as well as agents with negative initial emotions and different stances being simulated. Fig. 5 shows the views before and after the simulation within a limited time range. In general, the overall trend of emotional changes before and after the simulation remained consistent, but the intensity of negative group emotions (vertical axis) deepened, and the discussion heat (radius) also increased to a certain extent. In the early stage of the incident, opinion leaders called for protecting the doctor–patient relationship and improving the medical environment. Comparing the word cloud of this period with the word cloud of the same time period before the simulation, under the guidance of opinion leaders, the content of the word cloud changed, and the focus of group discussion shifted towards topics such as patient rights and the rational handling of doctor–patient relationships.

The emotions of the agents fluctuated continuously during the browsing process but generally remained in a relatively negative range. The stances of the agents changed many times during the simulation, but as the incident progressed towards the end, their stances mostly stabilized at neutral or slightly biased towards the hospital side. Notably, in Fig. 6, Agent No. 3 experienced a significant shift in stance in the early stage. Upon examining the browsing content of the agent at this point, it was found that the post condemned the program that published false reports from the perspective of news integrity. In the tree diagram, the browsing information of the agent at this turning point, along with the corresponding speech after browsing, can be traced, which helps to explain the agent's stance change and negative emotional shift.

In the early stages of the simulation, more opinion leaders supporting the patient's stance were introduced, and the results of the two simulations were compared and analyzed. There was some change in group emotion, but the magnitude was small; the more noticeable difference was in the distribution of the

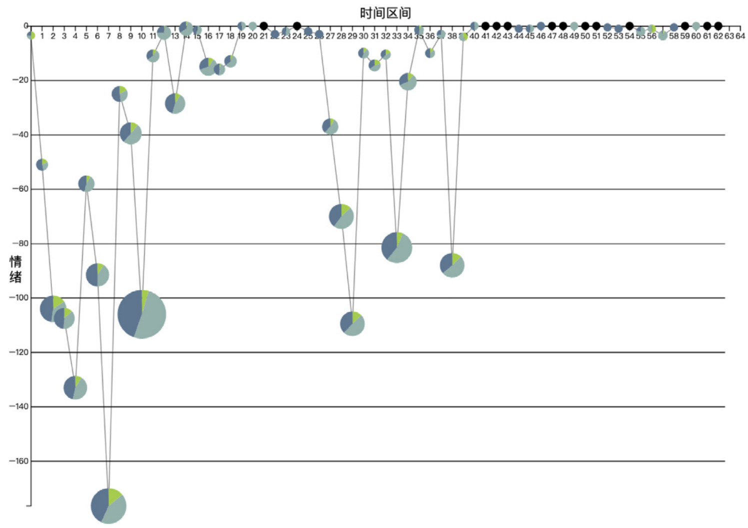

Fig. 4. Group stance and emotion.

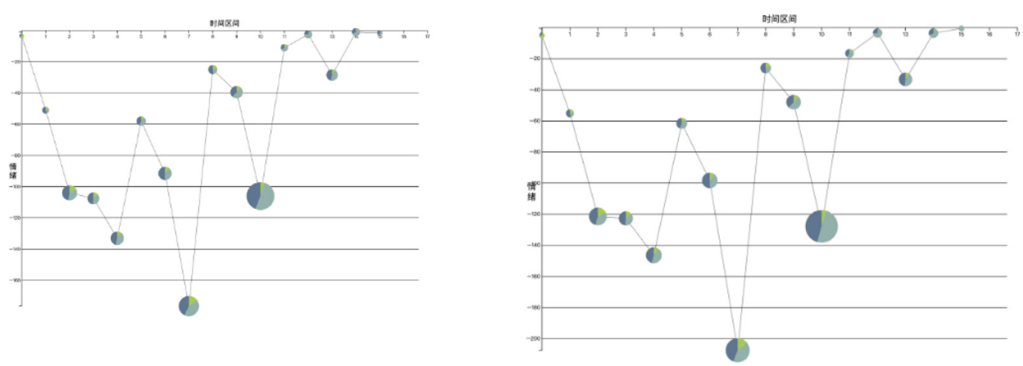

Fig. 5. Group stance and emotion comparison.

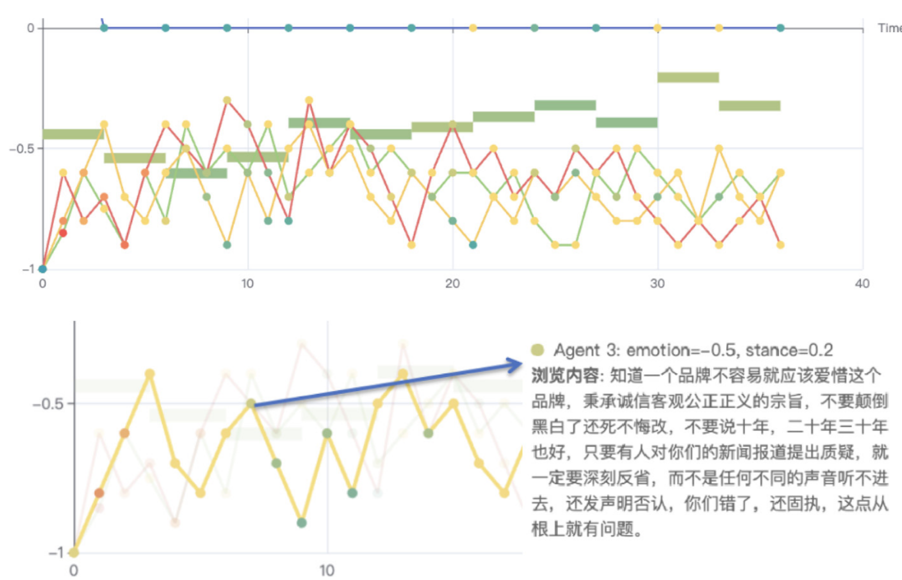

Fig. 6. Individual stance and emotion.

group's stance. During several time intervals with high discussion heat, the proportion of the group supporting the patient increased significantly. Fig. 7 also shows a clear bias towards the patient side within the agent group, as reflected by the newly appearing red and orange nodes that indicate more agents adopting the hospital stance. With the number of neutral agents decreasing, indicating the influence of opinion leaders' stance bias in social media on ordinary users.

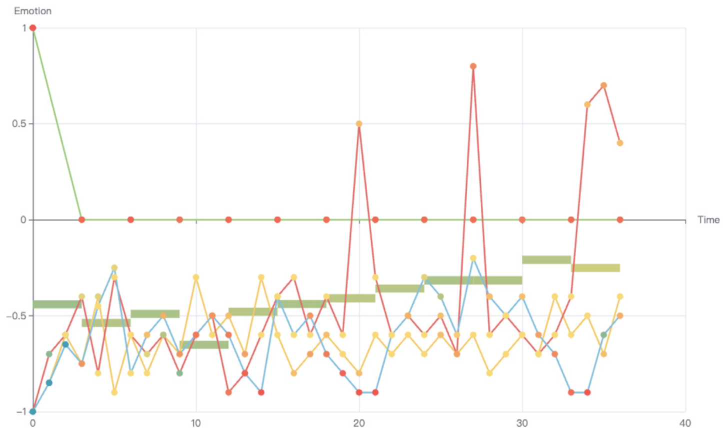

Fig. 7. Individual stance and emotion.

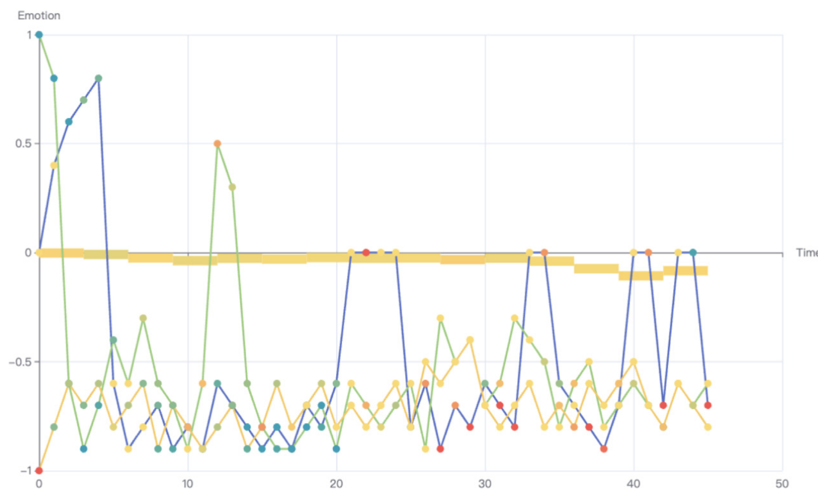

Fig. 8. Group stance and emotion.

6.3. Case 2: U.S. election

In the 2016 US election, Republican candidate Trump defeated Democratic candidate Hillary Clinton to become the 45th president of the United States. For this analysis, we crawled relevant Weibo posts on the Weibo platform from late November 2016 using the keyword "US presidential election". The same method was applied to process the dataset, which was then used for simulation analysis. The simulation selected all posts from November 27th to 28th as the basis, during which the Weibo platform discussed the recount of swing states and the controversy surrounding the election results. Ordinary users with neutral emotional stances, positive emotions supporting the Republican Party, and negative emotions supporting the Democratic Party were added to the environment.

From the analysis of the simulation results, it was found that ordinary users with different initial emotions tended to be more inclined towards negative emotions during the discussion. Users with unsteady personality characteristics (represented by the blue line in Fig. 8) demonstrated a tendency to be easily influenced by external information, causing their stances to fluctuate repeatedly. Other users remained relatively stable. As the simulation progressed, their stances eventually stabilized at neutral, similar to the behavior of rule-based agents. From a semantic perspective, comparing the word clouds before and after the simulation for this time period (Fig. 9), it was observed that, guided by the event overview set during the configuration phase, the agents engaged in more discussions about the topic of "recount" during the simulation.

7. Discussion

In the era of information, social media has become an important platform for the dissemination of public opinion. Its unique real-time and interactive features provide abundant data and context for studying the mechanisms behind public events. Based on a review of related research, we combine large language models, human–computer interaction, and visual analysis, and explore a simulation methodology driven by multiple agents.

Agents driven by large language models participate in discussions based on predefined role attributes. They dynamically adjust their stance and statements according to real data and generated data from other agents, providing a flexible and efficient method for simulating the evolution of public opinion events. By introducing personalized agent settings and interaction mechanisms, we optimize the efficiency and quality of data generation compared to traditional simulations. During the simulation, agents dynamically perceive event data and respond in real time, forming a complex social network. By integrating theories from traditional communication science and social psychology, we enhance the simulation of public opinion evolution and provide a novel tool for public opinion analysis and prediction in the context of social media.

Our system has broad application potential in practice. Data-driven analysis can provide insights and decision support for

Fig. 9. Word cloud before (left) and after (right) simulation, where public discussion shifts from candidates to re-ticketing.

public policy-making, public opinion management, and marketing. By simulating possible event development trends, the system can help predict key nodes and potential risk points in public opinion events, thereby supporting the formulation of response measures and guidance strategies.

There are still some limitations in our work: the content generated by large language models lacks the richness of semantic information compared to real social media content. To achieve better results, future work could involve fine-tuning the large language model with social media data.

We acknowledge that system response time and LLM inference latency may become bottlenecks when scaling to high (hundreds) agent counts. In future work, we plan to optimize the system pipeline with parallelized inference and caching strategies to ensure responsiveness and scalability in more demanding scenarios.

Currently, agent profiles focus on core psychological traits such as attitude firmness, while demographic attributes (e.g., gender, cultural background) and explicit social network structures are not modeled. Although this simplification facilitates computational feasibility, it limits the sociological realism of the simulation. Future work will integrate richer user profiling and social graphs to simulate phenomena such as homophily, community clustering, and cross-cultural discourse more accurately.

While our current evaluation focuses on trend-level alignment, future evaluation will emphasize micro-level behavioral validation. This includes tracking emotion propagation paths, quantifying the influence of key opinion leaders, and identifying conditions that trigger public opinion reversal. Such fine-grained assessment will strengthen the credibility and explanatory power of the simulation results.

Meanwhile a comprehensive evaluation requires user studies to assess the system's usability and effectiveness in realistic scenarios. Future work will incorporate systematic user assessments, focusing on participants' perception of simulation fidelity, interaction plausibility, and usefulness for understanding social media dynamics. Additionally, more detailed evaluations of interaction situations, such as content generation and event dissemination patterns, are essential to demonstrate the practical value of the simulation framework.

8. Conclusion

We propose an interactive visual analysis system that integrates agents into the context of social media. Building upon simulation implementation, the system leverages visualization techniques to enhance subsequent analysis and iterative exploration, introducing an interactive approach to social media event simulation and analysis. Through case validation on multiple datasets, the effectiveness and scalability of our system are demonstrated, proving the system's broad application potential in the research of public opinion events.

CRediT authorship contribution statement

Zichen Cheng: Conceptualization, Formal analysis, Methodology, Visualization, Writing – original draft, Writing – review & editing. **Ziyue Lin:** Conceptualization, Visualization. **Yihang Yang:** Conceptualization, Data curation, Project administration. **Zhongyu Wei:** Formal analysis, Project administration. **Siming Chen:** Conceptualization, Formal analysis, Project administration, Writing – review & editing.

Ethical approval

This study does not contain any studies with human or animal subjects performed by any of the authors. All data used in the study are taken from public databases that were published in the past.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

Example Prompt for Opinion Leader Agents:

At the current time step, the aggregated sentiment score is emo, where values closer to 1 indicate more positive emotion, and values closer to −1 indicate more negative emotion. The aggregated stance score is att, where values closer to 1 indicate stronger support for the hospital, and values closer to −1 indicate stronger support for the patient side.

As an opinion leader on the Weibo platform, your current stance is agent[att] and your emotion is agent[emo].

1. Based on the above information, generate a Weibo post that is natural and authentic in style, consistent with the identity of an opinion leader.

2. Update your internal emotion and stance after posting. The response format should be: 'Emotion: [Positive/Negative/Neutral], Stance: [Hospital/Patient/Neutral], Confidence: [percentage], Post: [content]'.

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